

Blinkit End-to-End Analytics using Microsoft Fabric

1. Bronze Layer

Purpose: The Bronze layer stores the raw ingested data exactly as received from the source system. It acts as the landing zone and preserves the original records for auditing and reprocessing.

```
SELECT * FROM bronze_orders;
SELECT * FROM bronze_products;
SELECT * FROM bronze_stores;
```

2. Silver Layer

Purpose: The Silver layer cleans, joins, and enriches the Bronze data. It calculates business fields such as order_amount and creates the target column is_late for analytics and machine learning.

```
CREATE OR REPLACE TABLE silver_orders AS
SELECT ...
(quantity * price) AS order_amount,
CASE WHEN delivery_minutes > promised_minutes
THEN 1 ELSE 0 END AS is_late;
```

3. Total Orders

Counts all orders available in the Silver layer. Used as a KPI card in Power BI.

```
SELECT COUNT(*) AS Total_Orders FROM silver_orders;
```

4. Total Revenue

Calculates the total sales revenue by summing the order amount.

```
SELECT SUM(order_amount) AS Total_Revenue FROM silver_orders;
```

5. Average Delivery Time

Finds the average delivery time of all completed orders.

```
SELECT AVG(delivery_minutes) AS Avg_Delivery_Time FROM silver_orders;
```

6. Revenue by Store

Shows which stores generate the highest revenue. Useful for comparing store performance.

```
SELECT store_name,
SUM(order_amount) AS Revenue
FROM silver_orders
GROUP BY store_name
ORDER BY Revenue DESC;
```

7. Revenue by City

Aggregates revenue city-wise to identify top-performing cities.

```
SELECT city,
SUM(order_amount) AS Revenue
FROM silver_orders
GROUP BY city
ORDER BY Revenue DESC;
```

8. Revenue by Category

Shows how much revenue each product category contributes.

```
SELECT category,
SUM(order_amount) AS Revenue
FROM silver_orders
GROUP BY category;
```

9. Orders by Hour

Counts orders for each hour to identify peak ordering periods.

```
SELECT order_hour,  
COUNT(*) AS Orders  
FROM silver_orders  
GROUP BY order_hour;
```

10. Top Products

Ranks products by revenue so the business can identify best-selling products.

```
SELECT product_name,  
SUM(order_amount) AS Revenue  
FROM silver_orders  
GROUP BY product_name  
ORDER BY Revenue DESC  
LIMIT 10;
```

11. Gold Layer

The Gold layer stores business-ready aggregated tables optimized for reporting and dashboards.

```
CREATE OR REPLACE TABLE gold_store_performance AS  
SELECT store_name,  
COUNT(*) AS Total_Orders,  
SUM(order_amount) AS Revenue,  
AVG(delivery_minutes) AS Avg_Delivery_Time,  
SUM(is_late) AS Late_Orders  
FROM silver_orders  
GROUP BY store_name;
```

12. Product Performance

Creates a summarized table for product-level analysis in Power BI.

```
CREATE OR REPLACE TABLE gold_product_performance AS  
SELECT product_name, category,  
SUM(quantity) AS Total_Quantity,  
SUM(order_amount) AS Revenue  
FROM silver_orders  
GROUP BY product_name, category;
```

13. Hourly Sales

Summarizes hourly revenue and order volume for trend analysis.

```
CREATE OR REPLACE TABLE gold_hourly_sales AS  
SELECT order_hour,  
COUNT(*) AS Orders,  
SUM(order_amount) AS Revenue  
FROM silver_orders  
GROUP BY order_hour;
```

14. Category Performance

Provides category-level KPIs such as revenue and average delivery time.

```
CREATE OR REPLACE TABLE gold_category_performance AS  
SELECT category,  
COUNT(*) AS Orders,  
SUM(order_amount) AS Revenue  
FROM silver_orders  
GROUP BY category;
```

15. City Performance

Aggregates city-level revenue, delivery performance, and late orders.

```
CREATE OR REPLACE TABLE gold_city_performance AS  
SELECT city,  
COUNT(*) AS Orders,  
SUM(order_amount) AS Revenue,  
SUM(is_late) AS Late_Orders
```

```
FROM silver_orders  
GROUP BY city;
```

16. Project Flow

Overall workflow: 1. Bronze - Raw Data 2. Silver - Cleaning & Feature Engineering 3. Machine Learning - Predict Late Deliveries 4. Gold - Business Aggregations 5. Power BI - Dashboards and KPIs